

Conditional diffusion emulator of Veros ACC equilibrium states: first end-to-end results

22 September 2026

Summary

- Goal: generate the post-spin-up (T, S) state of the Veros ACC channel directly from the EKE-closure coefficients (c_k, c_ϵ) , skipping spin-up.
- Pipeline: DINO-Fusion diffusion code adapted to a parameter ensemble; runs end to end on Jean Zay in under one A100 hour.
- Result: hold-out RMSE 0.061 / 0.076 / 0.141 K (scattered / band of three rows / top row) against the nearest training run 0.037 / 0.116 / 0.170 and the mean state 0.155 / 0.197 / 0.470 K: the model beats the nearest run in the two large-gap splits and loses only to the one-step neighbours of the scattered split. A Wasserstein metric on the stratification profiles gives the same ranking.
- The sampler’s land/padding constraint was the main error source: re-imposing exact zeros after every step (DINO’s constraint) instead of the noised fill the network saw in training cost 0.03–0.09 K (0.152 / 0.135 / 0.173 K before; same models, sampling only).
- Per-level min-max normalisation was tested and gives the same result as 3-std under the same sampler; dropped.
- Physical miss: the model generates temperature inversions at the rate of the weakly eddying runs in every regime.

1 Data

- 100 Veros ACC runs, 10×10 log-spaced grid: $c_k \in [0.0125, 0.8]$, $c_\epsilon \in [0.0875, 5.6]$; 100 years, 30-day output, interior grid $30 \times 42 \times 15$ (x, y, z).
- Kept: T and S , last 20 years (241 snapshots/run). $S = 35$ psu everywhere in every run.
- Nearly deterministic end states: within-window std 0.01 K above 1400 m vs 0.05 K between grid neighbours; bottom level still drifts by 0.45 K over the window.
- One effective parameter: $\log c_k - 1.05 \log c_\epsilon$ explains 99 % of the variance of the domain-mean T ; the low- c_k/c_ϵ part of the grid saturates.
- Per-level distribution of T (Figure 1): above 400 m a mixture of restoring values and interior, with σ_z set by the meridional gradient, so $\mu_z \pm 3\sigma_z$ is wider than the level and the normalised values stay within $[-0.85, 0.6]$ ($[-0.7, 0.4]$ in the top 200 m); below 1400 m a warm tail to $x' = +2.7$. Run-to-run signal σ_z^{runs} : 0.7 % of $3\sigma_z$ at the surface, 26 % at the bottom.

2 Method

- Fields $\rightarrow$ 30 channels $(T_{1..15}, S_{1..15})$, normalised per level, land $\rightarrow 0$, zero-padded $42 \times 30 \rightarrow 48 \times 32$:

$$x'_z = \frac{x_z - \mu_z}{3 \max(\sigma_z, 0.05)}. \quad (1)$$

μ_z, σ_z are computed once on all 100 runs and held fixed for every hold-out split (a deliberate, mild leakage of 30 scaling constants; no held-out state enters training). The sampler clips the predicted clean state at $|x'| \leq 1$ (three standard deviations) at every step.

Distribution of T per level, all 100 runs $\times$ 241 snapshots, water cells (surface at the top). Black: μ_z ; red dashed: $\mu_z \pm 3\sigma_z$; orange ticks: the 100 per-run level means.

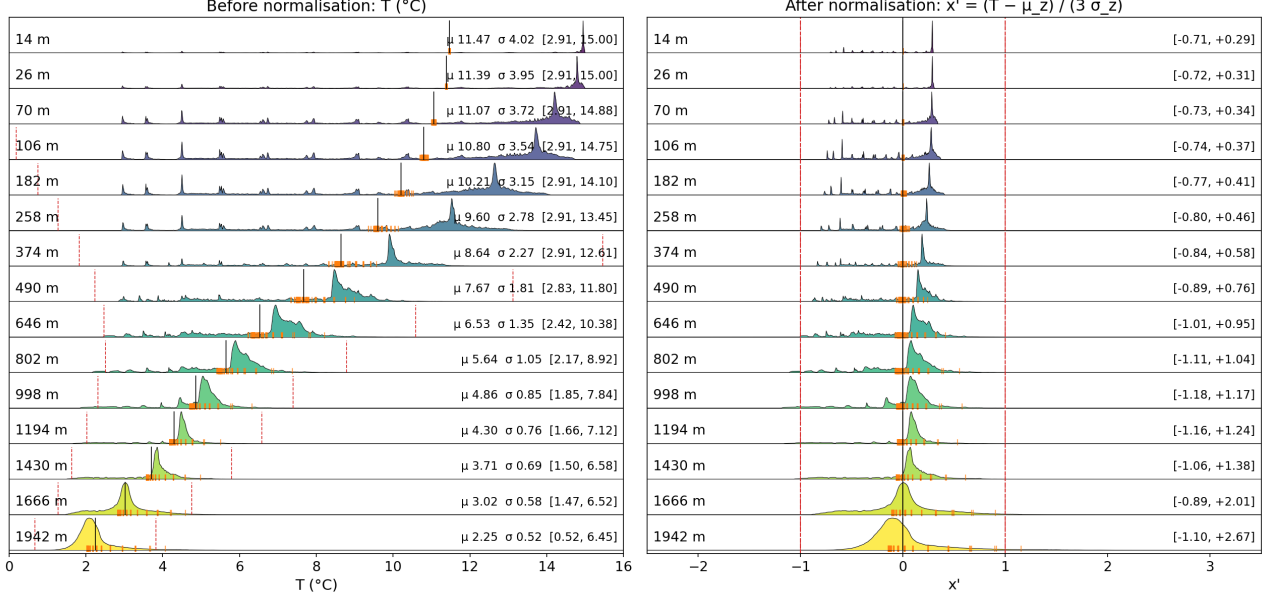


Figure 1: Distribution of T per level over all 100 runs and 241 snapshots (water cells), before (left) and after (right) the per-level normalisation of Eq. (1). Black: μ_z ; red dashed: $\mu_z \pm 3\sigma_z$; orange ticks: the 100 per-run level means.

- Denoiser: UNet2DModel (64, 64, 128, 128), 6.9 M parameters. The condition c , the standardised pair $(\log c_k, \log c_\epsilon)$, enters through an MLP added to the timestep embedding.
- DDPM, ϵ -prediction, 1000 steps, squared-cosine schedule, EMA:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \epsilon} \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, c) \right\|^2. \quad (2)$$

20 000 steps, batch 32, AdamW 10^{-4} with cosine decay; 22 min on one A100.

- Sampling: 1000-step DDPM loop with EMA weights; after every step the land and padding cells are reset to the fill at the noise level of that step, $\sqrt{\bar{\alpha}_t} f + \sqrt{1 - \bar{\alpha}_t} z$ (here $f = 0$), as those cells looked in training; resetting them to the exact fill (DINO’s BorderZeroConstraint) is the ablation of Table 1. 32 samples per condition.

3 Evaluation

- Truth: 20-year time-mean state of each held-out run; metrics on water cells.
- Baselines: mean training state; nearest training run in standardised log-parameter space; average of the axis-neighbour training runs on the grid.
- Three hold-out splits, same data file and statistics, different training sets: scattered (10 interior points), band of three rows ($c_k = 0.126, 0.2, 0.3175$; 30 runs; training rows a factor 6.3 apart), top row ($c_k = 0.8$; 10 runs; extrapolation).
- Distributional metric. Each state is reduced to its profile of horizontal means over water cells, $\bar{T}_z$. Per level, the Wasserstein-1 distance between the 32 generated values and all 241 true snapshots of the window, through the quantile functions,

$$W_1^{(z)} = \int_0^1 |F_{g,z}^{-1}(q) - F_{t,z}^{-1}(q)| dq, \quad W_1 = \sum_z \frac{\Delta z_z}{H} W_1^{(z)}, \quad (3)$$

with Δz_z the level thicknesses from the level midpoints ($H = 2080$ m). All 241 snapshots are used, not a subsample of 32: the quantile form accepts unequal sizes and a subsample would only add noise. A point prediction x (the baselines) gives $W_1^{(z)} = \frac{1}{241} \sum_i |x_z - \bar{T}_{z,i}|$. Unlike the RMSE of the

ensemble mean, this rewards a correct spread and does not reward collapse to the mean; it ignores horizontal structure, which the RMSE keeps.

	scattered 10 runs	band of three rows 30 runs	top row 10 runs
diffusion, ensemble mean of 32: RMSE (K)	0.061	0.076	0.141
diffusion, single sample: RMSE (K)	0.147	0.160	0.208
nearest training run: RMSE (K)	0.037	0.116	0.170
neighbour average: RMSE (K)	0.020	0.116	0.170
mean training state: RMSE (K)	0.155	0.197	0.470
diffusion: domain-mean bias (K)	+0.042	+0.020	−0.016
ensemble spread / true spread (K)	0.128 / 0.031	0.132 / 0.031	0.132 / 0.032
T inversions (interfaces), generated / truth	10.2 % / 7.7 %	10.9 % / 1.3 %	12.3 % / 0.1 %
max salinity error (psu)	0.013	0.014	0.017
W_1 profile, diffusion (K)	0.066	0.075	0.132
W_1 profile, neighbour average (K)	0.036	0.095	0.142
W_1 profile, nearest training run (K)	0.044	0.095	0.142
W_1 profile, mean training state (K)	0.136	0.160	0.420
same models, exact fill (previous sampler): RMSE (K)	0.152	0.135	0.173
same models, exact fill (previous sampler): W_1 (K)	0.127	0.111	0.143

Table 1: Hold-out metrics per split, mean over the held-out runs, water cells only, 32 samples per run; last two rows: the same models with the previous sampler (exact zeros on land and padding after every step).

4 Findings

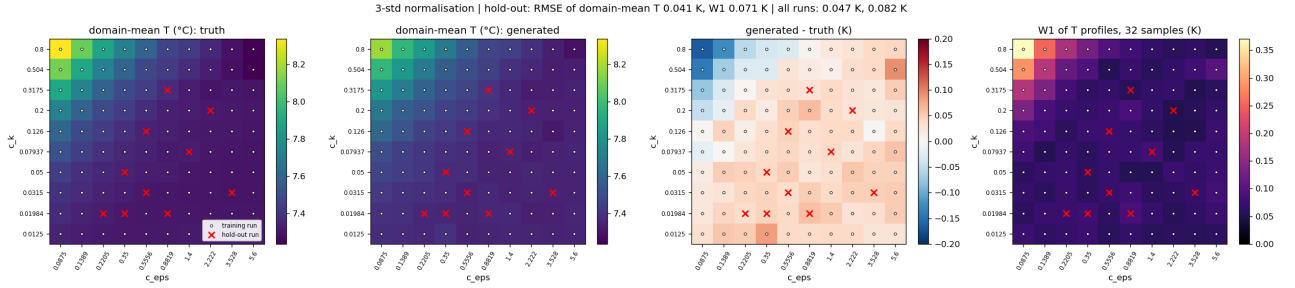
- **The land/padding constraint was the main error source.** Same models, sampling only: exact zeros re-imposed after every step give 0.152 / 0.135 / 0.173 K, zeros re-imposed at the step’s noise level give 0.061 / 0.076 / 0.141 K. In training those cells carry noise like every other; at high noise levels the network had never seen a solid block of zeros around the domain. The gain is largest at the bottom ($0.34 \rightarrow 0.07$ K, scattered) and the earlier mean-state collapse was this artefact: the model now sits at 40 % of the mean-state error in the scattered split, and W_1 at the training grid points drops from 0.124 to 0.083 K.
- **Model vs baselines.** Band split: beats the nearest run at every level below 26 m and in all three rows (0.063 / 0.074 / 0.091 K vs 0.063 / 0.132 / 0.152 K for $c_k = 0.126 / 0.2 / 0.3175$). Top row: beats the nearest run at every level but the surface and the bottom, and copying the row below for c_e 0.14–1.4 (0.05–0.26 vs 0.08–0.31 K); loses at the corner (0.41 vs 0.35 K) and in the flat regime ($c_e \geq 3.5$: 0.09–0.10 vs 0.03–0.08 K). Scattered split: loses to the one-step neighbours (0.061 vs 0.037 / 0.020 K) except between 490 and 650 m (Figures 2, 3, 4).
- **Depth structure.** Error 0.04–0.08 K at every level in the scattered split; the two bottom levels are no longer special (0.07 K) except in the top row (0.29 K, driven by the corner runs). Worst band: the top 100 m (scale $3\sigma_z \approx 12$ K amplifies sampler noise).
- **The sampler’s clip is a regulariser, not a range limit.** Same trained models, sampling only (exact fill): clip at $|x'| \leq 1$ gives 0.152 / 0.173 K (scattered / top row), at 1.5 it gives 0.194 / 0.183 K, at 3 it gives 0.302 / 0.277 K, although the final samples barely exceed $|x'| = 1$ (0.1 % of values): clipping the early, inaccurate clean-state estimates keeps the chain on track. The price of the clip is a range limit for extrapolation: with the fixed statistics 12 % of the held-out top row’s true bottom values lie above $\mu_z + 3\sigma_z$ and cannot be generated; the corner run is the top row’s largest error in every variant.
- **Inversions.** Generated 10–12 % of interfaces with $\partial_z T < 0$ in every regime; truth: 7.7 % (scattered) $\rightarrow$ 1.3 % (band) $\rightarrow$ 0.1 % (top). Unchanged by the sampler: not learned.
- **W_1 on the mean profiles agrees with the RMSE ranking** (right-hand panels of Figures 2, 3, 4): training grid points score like the held-out ones (0.083 vs 0.071 K, scattered), and from 490 m down the samples beat the point baselines in the band and top splits (0.058 vs 0.115 K at 650 m, band), because a point prediction is charged for the spread it cannot represent.

- **Plumbing.** S within 0.02 psu of 35 with no constraint; land exact; ensemble spread $4\times$ the true within-window spread (Figures 5, 6, 7).

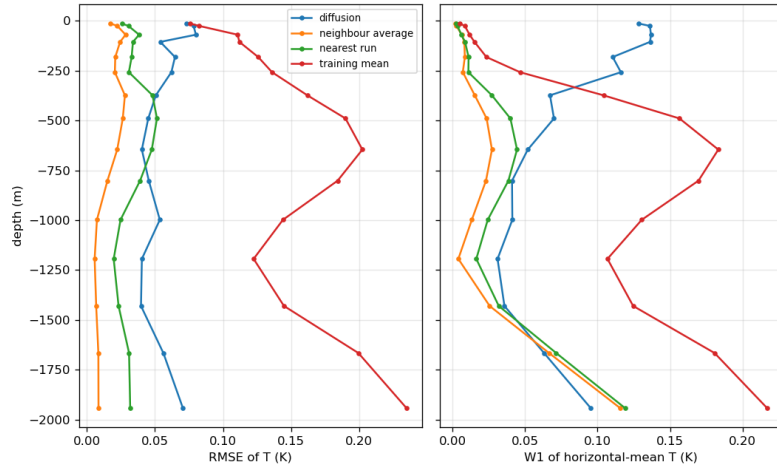
5 Next steps

- Stratification constraint at sampling time (DINO-Fusion’s isotonic projection, on T since S is constant): targets the inversion miss directly.
- Top-row corner: the clip’s range limit is now the dominant extrapolation error; a clip that follows the conditioning (bounds from the nearest training rows) is the next sampling-only test.
- Stronger conditioning: classifier-free guidance; year as a third condition (drift of 0.45 K at the bottom inside the window). A second training seed to size the seed noise.
- Evaluate against snapshots as well as the time mean (nearest-snapshot RMSE, spread–skill).
- Then: same pipeline on a global Veros configuration (active S , seasonal cycle, multi-century spin-up).

Reproducibility. Branch `miniveros` of the DINO-Fusion repository, package `miniveros/`. Training at `2b542aa`; sampling and evaluation at `bfee32f` (`fill_mode` noised; the exact-zero rows: clean). Extraction 2.5 min (8 cores), training 17–22 min (one A100), generation of the hold-out and grid sets + evaluation 14–18 min.

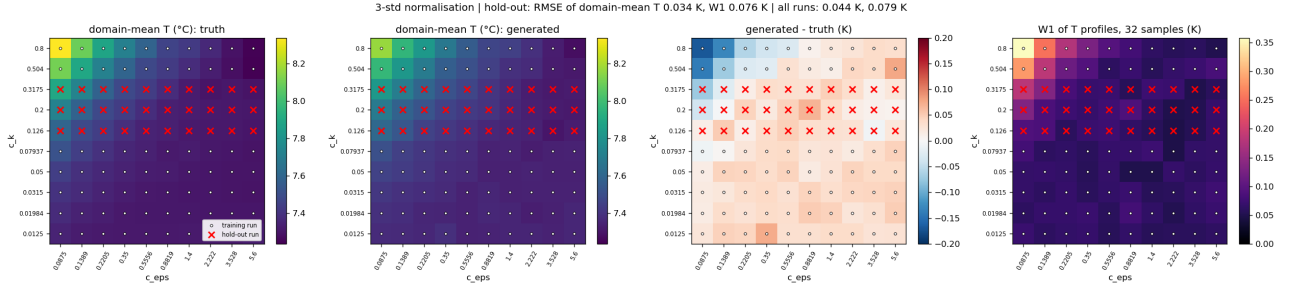


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

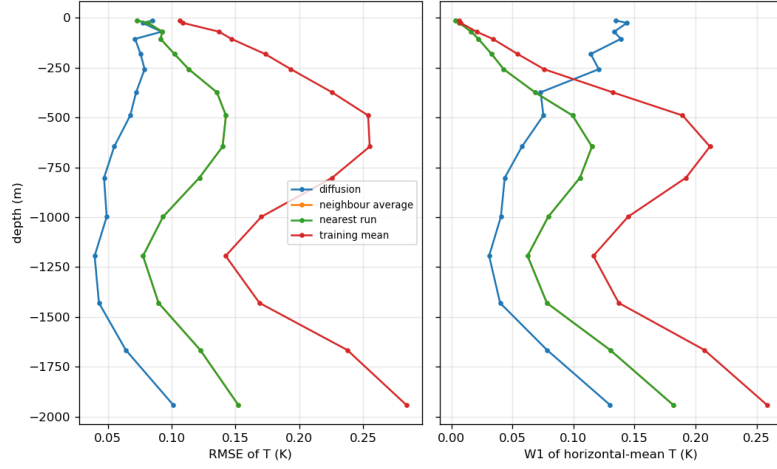


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 2: Scattered split (10 interior held-out runs), 32 samples per run.

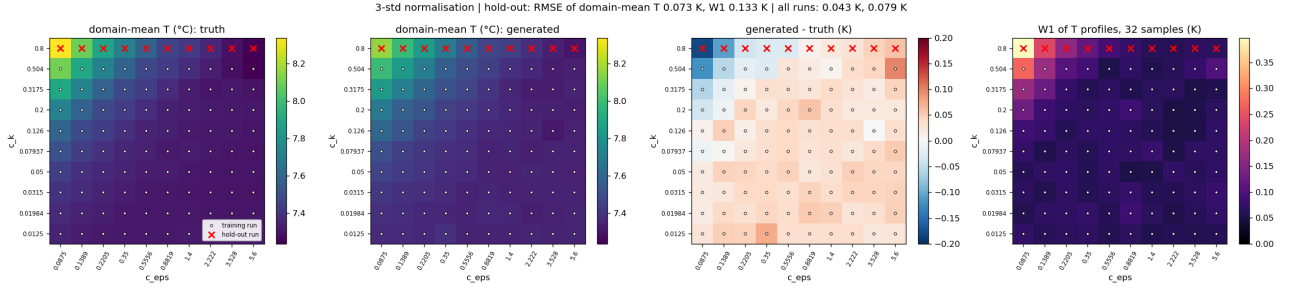


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

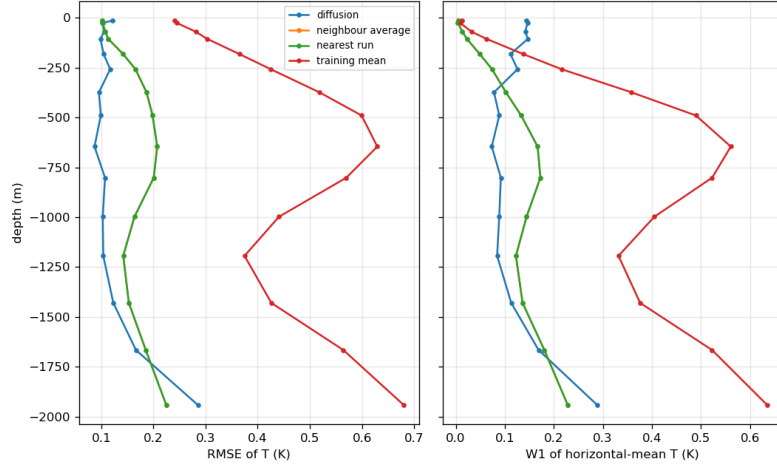


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 3: Band of three rows ($c_k = 0.126, 0.2, 0.3175$ held out), 32 samples per run.



(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.



(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 4: Top row ($c_k = 0.8$ held out, extrapolation), 32 samples per run.

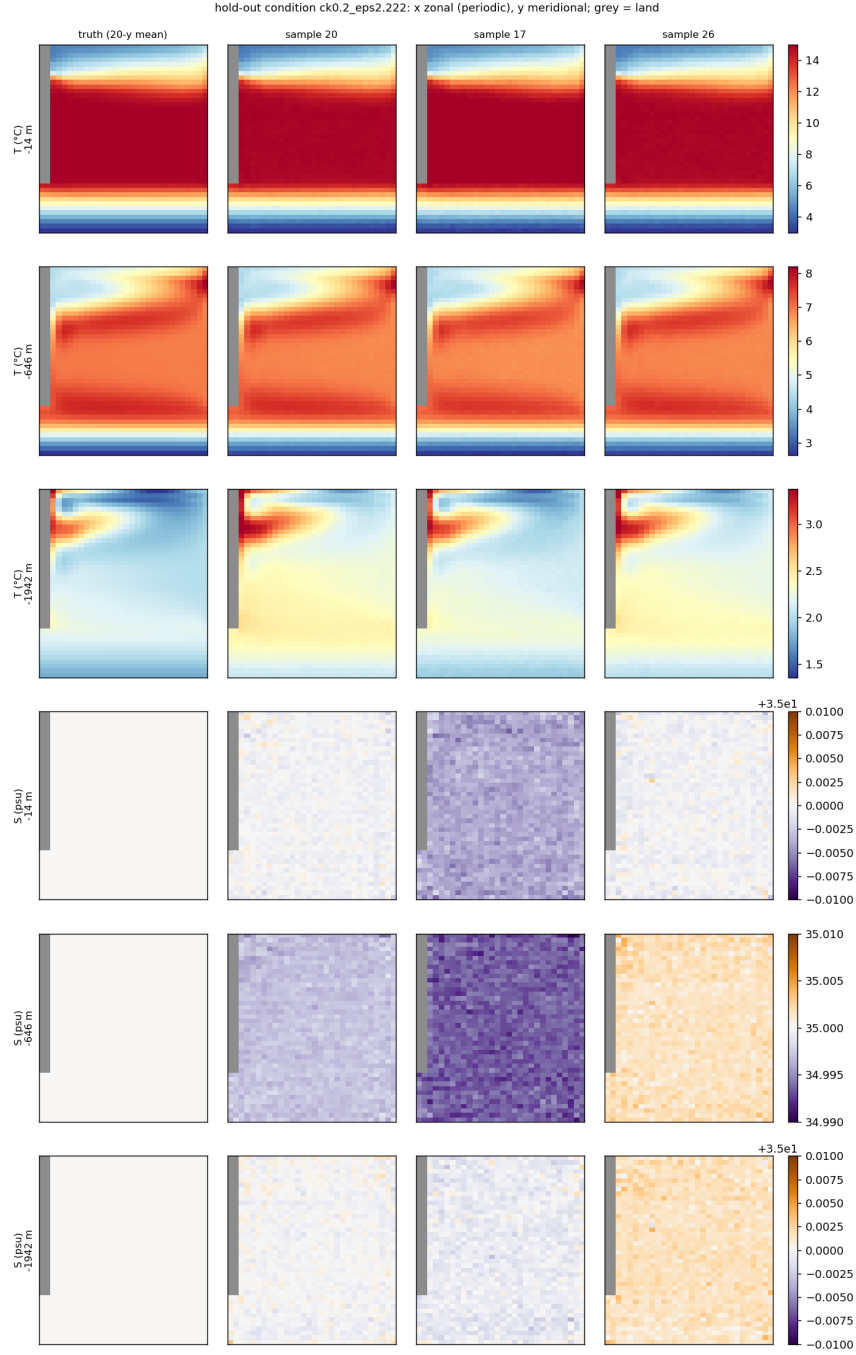


Figure 5: Scattered split, $c_k = 0.2$, $c_\epsilon = 2.222$: true 20-year mean (left) and three random samples at the surface, 650 m and the bottom; T (top three rows), S (bottom three rows). Grey: land.

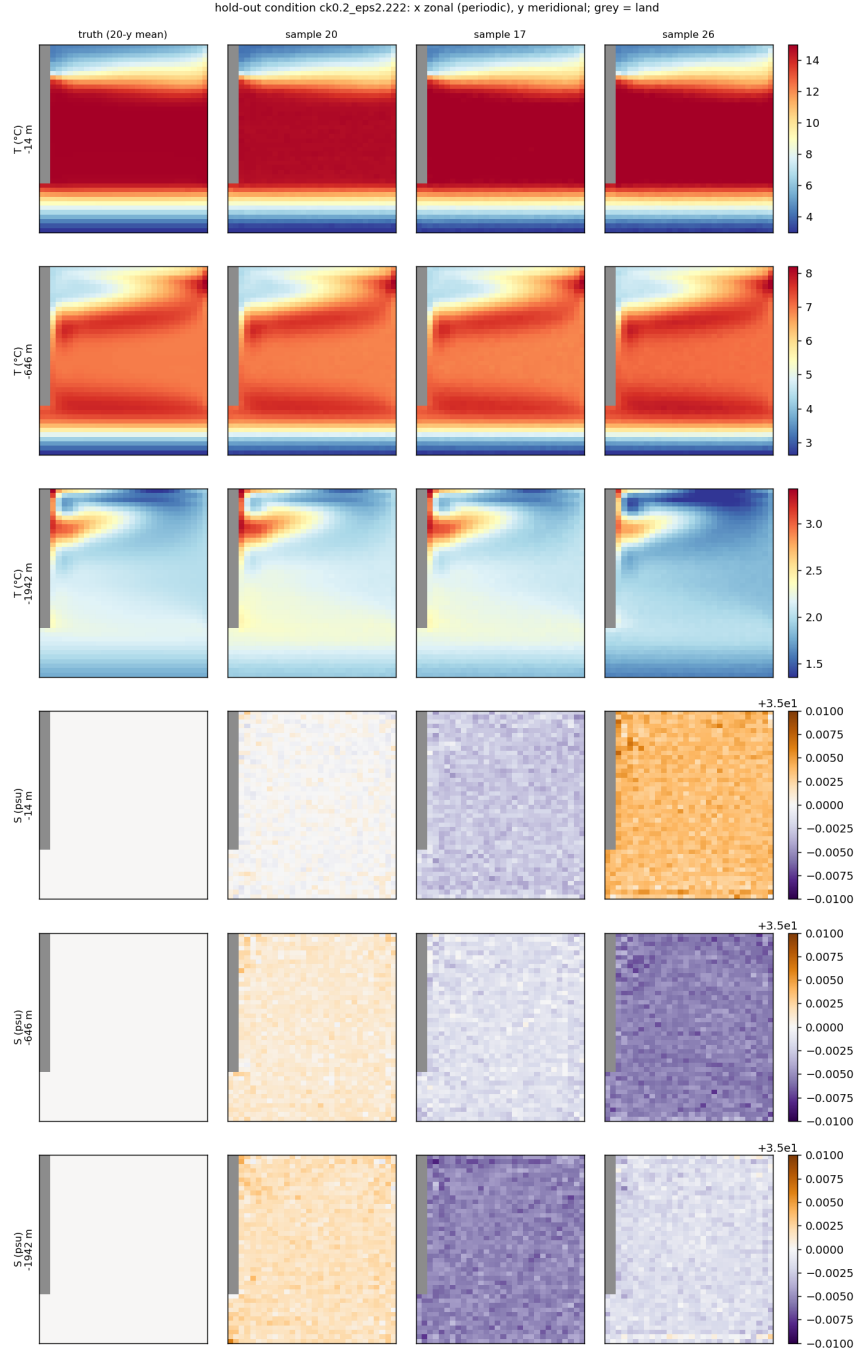


Figure 6: Band-of-three split, same condition ($c_k = 0.2$, $c_\epsilon = 2.222$, now in the middle held-out row).

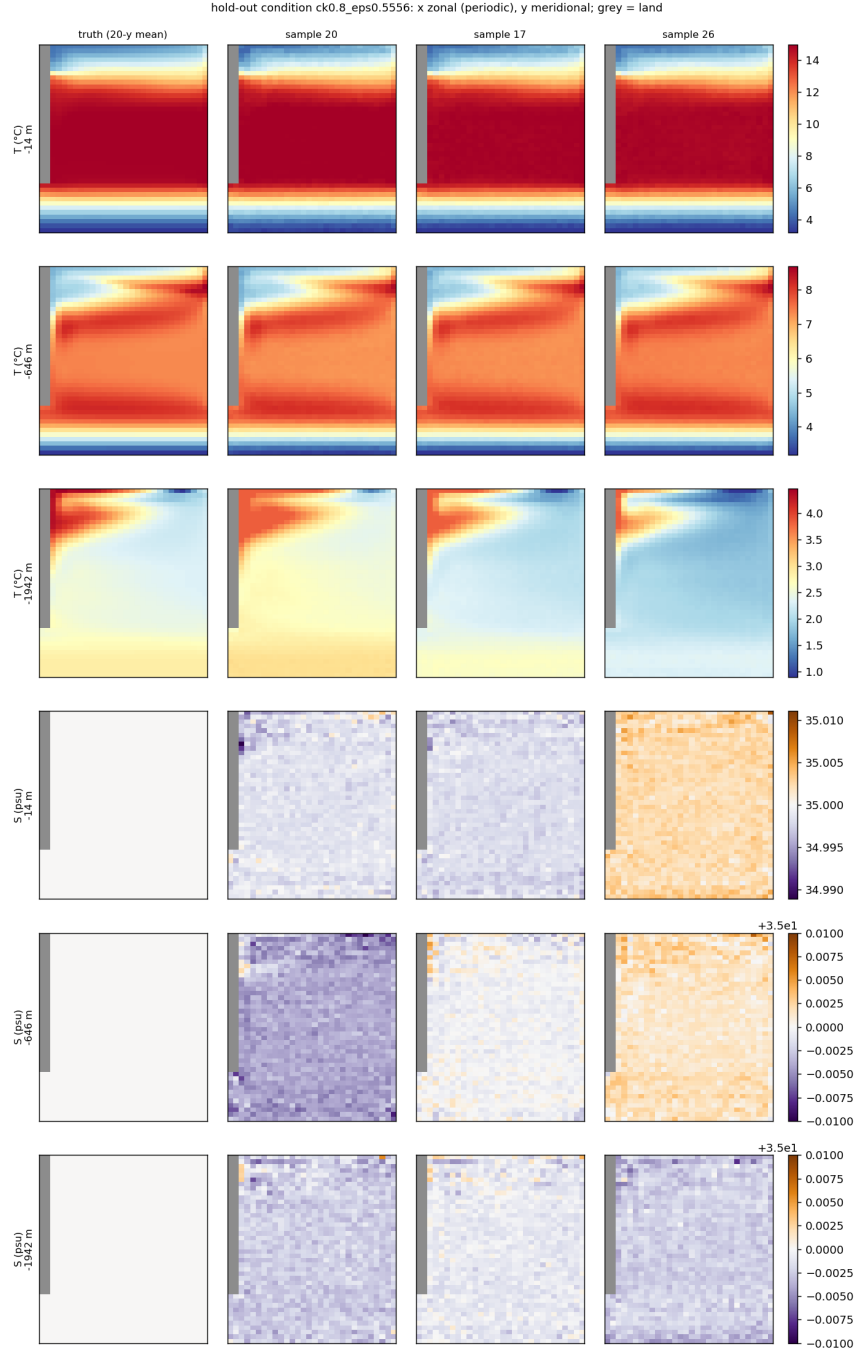


Figure 7: Top-row split, $c_k = 0.8$, $c_\epsilon = 0.5556$ (extrapolation one row above the training grid).